



Playing Hide and Seek with Passwords

Passwords are used in many ways for security in computers, from BIOS management to logging in to your user account; from accessing hard disk contents, to restricting access to specific services. For Linux, the root password is the most important; if you have it, you “own” the machine. This article is like a drill, where we break a password, find a way to protect it, and again break it. Who wins? Read on to find out...

We are going to be breaking the *root* password on an RHEL6 machine. We start with switching to single-user mode. Here are the steps:

1. Press any key as soon as the booting starts. This will display the GRUB menu.
2. Now press *e* to edit, and go down to the second line, which begins with *kernel*.
3. Again press *e* to edit kernel parameters, and at the end of the line, just add a *1* (see Figure 1) to start the system in single-user mode instead of the default run-level.
4. Press *Enter* to quit kernel parameter editing mode, and then press *b* to boot the modified options.
5. The system boots to single-user mode; now we just change the *root* password with the *passwd* command. If the command doesn't display any output, it indicates that SELinux is in enforcing mode, so we will have to disable it using the *setenforce 0* command. Then again run *passwd* and change the password. After you have changed the password, you can switch to graphical

mode using the *init 5* command and login to the *root* account in a terminal, using the newly assigned password.

It's easy to change the *root* password, isn't it? But this can become a nightmare if someone else does this to your system. So we try to block this approach. Notice that we got access to single-user mode without any password; to patch this hole, we shall require a password to access single-user mode. In RHEL5 you need to write *su:S:wait:/sbin/sulogin* in */etc/inittab* file before *id:5:inittab*. But this doesn't work in RHEL6. So we edit the */etc/rc1.d/S99single* file and write *exec /sbin/sulogin* before *exec init -t1 S*. Now if you try to switch into single-user mode, you'll have to enter the *root* password.

OK—can we relax, now that single-user mode is password-protected?

No—the game has just begun, and will get more interesting. Here is the next twist: you can *still* change the *root* password. In our first go, we edited the kernel